

When the Inverted-U Dissolves: Structural Boundary Conditions of the Internationalization–Performance Relationship in Pacific Microstates

Keywords: internationalization–performance relationship; inverted-U; boundary conditions; small island developing states; forced internationalization; institutional voids; labor productivity; Pacific economies

1. Introduction

The relationship between internationalization and firm performance (I–P) is among the most examined—and most contested—regularities in international business (IB) research (Verbeke & Brugman, 2018). Decades of work have converged on a non-linear, inverted-U functional form: foreign expansion first improves performance through scale economies, learning, and market diversification, then erodes it beyond a turning point as coordination costs and the liability of foreignness accumulate (Contractor, Kundu, & Hsu, 2003; Hitt, Hoskisson, & Kim, 1997; Lu & Beamish, 2004). Meta-analyses report a positive average effect with very large heterogeneity, an invitation to ask not *whether* internationalization helps but *under what conditions* (Bausch & Krist, 2007; Kirca et al., 2011).

A foundational—but rarely stated—assumption underlies this entire tradition: that the institutional and market environment may shift the *intensity* or the *turning point* of the I–P relationship, but not its fundamental shape. Recent large-sample replications have begun to punc-

ture that assumption, finding the multinationality–performance relationship to be weak, unstable, or context-dependent across countries (Berry & Kaul, 2016; Pisani, Garcia-Bernardo, & Heemskerk, 2020). Yet the boundary of the inverted-U paradigm—the conditions under which it ceases to hold at all—has not been mapped at the firm level in the contexts where those conditions are most plausibly absent.

This paper maps that boundary. We argue that the inverted-U logic rests on three structural prerequisites that are simultaneously violated in the world’s most peripheral economies—the Pacific small island developing states (SIDS): (i) a viable domestic market large enough for firms to attain minimum efficient scale before exporting; (ii) trade costs low enough for export expansion to be an economic optimization rather than a survival imperative; and (iii) institutional support sufficient for firms to appropriate returns from international activity (Briguglio, 1995; Khanna & Palepu, 2010; Winters & Martins, 2004). Where all three fail, firms export not by strategic selection but by compulsion, and the cost–benefit machinery that generates the inverted-U has no purchase.

Using World Bank Enterprise Survey (WBES) microdata for 1,450 firms across seven Pacific SIDS, we ask whether the inverted-U extends to this extreme periphery. It does not: the curvature that defines the inverted-U is statistically absent and the export-intensity slope is indistinguishable from zero. A weaker, suggestive pattern—conditional on firm capability, the relationship leans convex rather than concave, a productivity penalty in the low-to-moderate export range in which nearly all island firms operate—points toward a theoretical limiting case we name the **Forced Internationalization Penalty (FIP)**, which we treat as a bounding concept rather than a confirmed effect.

We make four contributions. First, we provide the first firm-level test of the *scope conditions* of the inverted-U paradigm in Pacific microstates, showing that the dominant functional form

does not generalize to the extreme periphery. Second, we make explicit the three structural prerequisites that the I–P literature has left implicit—a theoretical clarification applicable to any micro or peripheral economy. Third, we introduce and delimit FIP as a theoretical construct, distinguishing it from phase-one S-curve costs and the liability of foreignness. Fourth, we draw out the policy implication that export-led-growth prescriptions are misaligned with the structural realities of microstates, where the binding constraints are domestic-market viability, trade costs, and institutions rather than export intensity *per se*.

2. Theory and Propositions

2.1 The inverted-U and its implicit prerequisites

The inverted-U derives from a cost–benefit logic (Hitt et al., 1997; Lu & Beamish, 2004). Early internationalization yields scale economies, scope for learning, and risk diversification; beyond a threshold, organizational complexity, coordination costs, and the liability of foreignness reverse the trajectory (Hennart, 2007; Zaheer, 1995). The turning point—typically estimated at 30–70% of foreign sales to total sales (FSTS)—is the intensity at which marginal cost equals marginal benefit.

This machinery presupposes that exporting is a *choice* made from a viable home base. Three structural prerequisites are required for that presupposition to hold:

Prerequisite 1 — a viable domestic market. The Uppsala account assumes a functioning home base that supplies revenue, resources, and learning before and during expansion (Johanson & Vahlne, 1977, 2009). When the domestic market cannot support minimum efficient scale, firms are compelled to export regardless of readiness, and the “selection into exporting” that underlies most I–P models breaks down (Melitz, 2003).

Prerequisite 2 — manageable trade costs. The model assumes that the marginal cost of serving foreign markets is bounded. Extreme geographic isolation in the Pacific inflates logistics and freight costs by wide margins, so that export expansion erodes margins at every intensity (Briguglio, 1995; Winters & Martins, 2004).

Prerequisite 3 — functional institutional support. Appropriating returns from internationalization requires intermediary institutions—trade finance, logistics, export-promotion capacity—whose absence (institutional voids) imposes a structural drag (Doh, Rodrigues, Saka-Helmhout, & Makhija, 2017; Khanna & Palepu, 2010; North, 1990).

2.2 From dissolution to inversion

When all three prerequisites fail simultaneously, the inverted-U’s generating mechanism is disabled. We therefore expect, first, that the *shape* itself fails to appear:

Proposition 1 (dissolution). In Pacific SIDS, the inverted-U is absent: neither the FSTS curvature term nor the FSTS slope is statistically distinguishable from zero, unlike in mainland economies where the prerequisites hold.

Dissolution is the minimal claim. A stronger possibility is that, once firm-level capability differences are accounted for, the relationship turns mildly *convex*. If exporting is compulsory and prerequisite-deprived, additional export intensity in the empirically dominant low-to-moderate range raises exposure to high trade costs and coordination burdens without a domestic base to absorb them—a penalty—whereas only the rare, highly capable, high-intensity exporter escapes it. This implies a convex (U-shaped), not concave, conditional relationship:

Proposition 2 (conditional convexity; exploratory). Conditional on firm capability, the SIDS I–P relationship leans convex rather than concave—a productivity

penalty across the low-to-moderate export-intensity range, reversing only at high intensities that are empirically rare. We frame this as an exploratory proposition, given that the convexity is only marginally significant.

Capability is a level-shifter, not a rescue. Absorptive-capacity and resource-based logic implies that technological capability raises productivity at every intensity (Cohen & Levinthal, 1990; Lall, 1992), but it cannot supply the missing structural prerequisites:

Proposition 3 (capability does not restore the curve). Technological capability and digital adoption raise the productivity level but do not restore a concave (inverted-U) relationship.

2.3 The Forced Internationalization Penalty as a theoretical limiting case

We name the limiting case of Propositions 1–2—where compulsory, prerequisite-deprived internationalization becomes monotonically performance-eroding—the **Forced Internationalization Penalty (FIP)**. FIP is distinct from adjacent constructs. Unlike the downward phase-one slope of three-stage theory (Contractor et al., 2003), FIP is not a transient learning cost awaiting an eventual turning point; it is a structural condition with no interior optimum in the observable range. Unlike the liability of foreignness, which operates at the host-market frontier (Zaheer, 1995), FIP originates at the home (source) base. And unlike necessity entrepreneurship, where founders in developing contexts can attain parity given time and institutional support (Sarasvathy, Kumar, York, & Bhagavatula, 2014), FIP characterizes contexts where that support is systematically absent. We treat FIP as a *theoretical* construct delimiting the scope of universal internationalization theories; the empirical SIDS evidence below is consistent with its direction but, as we report transparently, does not robustly estimate its magnitude.

3. Data and Methods

3.1 Sample

We draw firm-level data from the World Bank Enterprise Survey for the seven Pacific small island developing states with available firm microdata—Fiji, Kiribati, Papua New Guinea, Samoa, Solomon Islands, Tonga, and Vanuatu—across all available waves from 2009 to 2025. These are Pacific Island Countries under the World Bank’s regional classification and members of the United Nations’ Pacific SIDS list. We deliberately exclude Timor-Leste, which—although occasionally grouped with the region—is neither a World Bank Pacific Island Country nor a United Nations Pacific SIDS, and is classed as a separate East Asia and Pacific economy; including it would conflate a structurally distinct, comparatively large economy (it would contribute nearly a quarter of the pooled observations) with the genuine island periphery. After requiring non-missing labor productivity and export intensity, the analytic sample comprises **1,450 firms across seven economies and thirteen country-year cells**. Consistent with the structural argument, the export profile is exceptionally thin: only **14.6%** of firms export, mean FSTS is **0.054**, and within-economy productivity dispersion is high (the 90th/10th-percentile labor-productivity ratio exceeds an order of magnitude), consistent with the resource-misallocation signature of weak-institution contexts (Hsieh & Klenow, 2009).

3.2 Variables

Labor productivity is the natural log of annual sales per worker, $\ln(d2/11)$, winsorized 1/99 and standardized (z-scored) within each economy-year cell so that estimates are currency-neutral and on the same footing as the companion 49-economy study. Internationalization intensity (FSTS) is total exports as a share of sales, $(d3b + d3c)/100$, centered within sample. Technological capability (TCI) is a standardized index of quality certification and foreign-technology licens-

ing; digital adoption (DAI) is the website-presence indicator—the Tier-1 measure comparably available across SIDS waves. Firm age (log years since establishment) and foreign ownership (a 10%-threshold indicator) are controls. We follow established practice in operationalizing capability as a formative composite (Diamantopoulos & Winklhofer, 2001; Jarvis, MacKenzie, & Podsakoff, 2003).

3.3 Estimation and inference

All models include economy and year fixed effects, isolating within-economy-year variation. Because the design has only **seven clusters (economies)**, conventional cluster-robust (CRV1) standard errors over-reject; we therefore base inference on the **wild-cluster restricted bootstrap** (Cameron, Gelbach, & Miller, 2008; 9,999 replications, Rademacher weights, clustered by economy) and report cluster counts prominently. We test the inverted-U using the conditions of Haans, Pieters, and He (2016) and the Lind and Mehlum (2010) U-test, and we contrast the SIDS estimates with the sharp mainland inverted-U documented for transition economies in the companion 49-economy study.

4. Results

4.1 The inverted-U dissolves (Proposition 1)

Table 1 reports the model ladder. Adding a linear FSTS term to the controls leaves the slope indistinguishable from zero (M1: $\beta = -0.085$, wild-bootstrap $p = .66$). Adding the quadratic term does not recover an inverted-U: the curvature coefficient is **positive** and only marginally significant (M2: $\beta_2 = +0.696$, wild-bootstrap $p = .082$; $\beta_1 = -0.567$, $p = .088$). The Lind–Mehlum conditions for a genuine inverted-U are not satisfied—there is no interior maximum in the observable range. This stands in sharp contrast to the mainland transition regime, where the

same specification yields a sharp, highly significant inverted-U ($\beta_2 = -1.012$, $p < .001$). The canonical shape is therefore **absent at the periphery**.

Table 1. Internationalization and standardized labor productivity, Pacific SIDS (economy + year FE; firm-age and foreign-ownership controls; wild-cluster bootstrap p, $G = 7$).

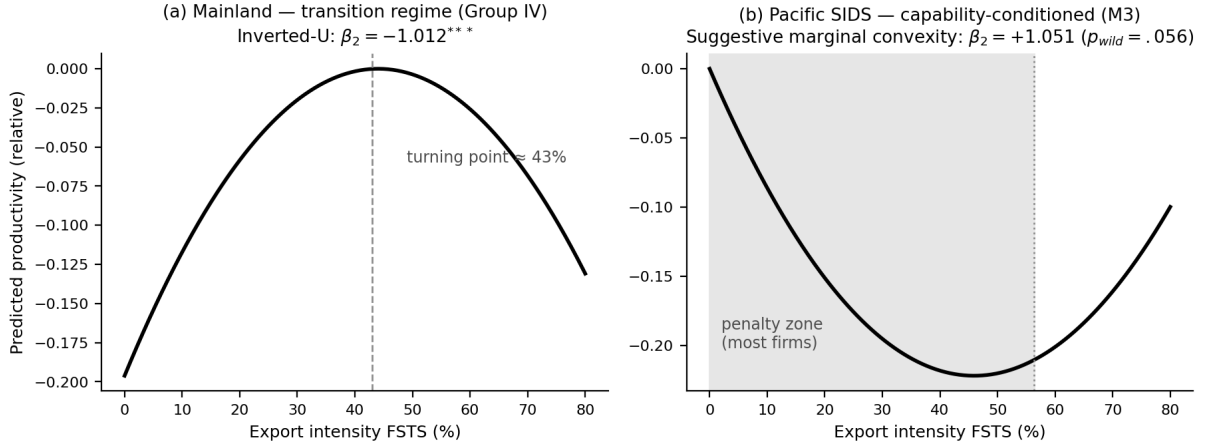
Term	M1	M2	M3
FSTS (centered)	-0.085 (.66)	-0.567 (.088)	-0.852 (.024)
FSTS ²	—	+0.696 (.082)	+1.051 (.056)
TCI (z)	—	—	+0.064 (.036)
DAI (website)	—	—	+0.157 (.102)
N	1,434	1,434	1,389

Dependent variable: labor productivity z-scored within economy-year. Firm-age and foreign-ownership controls and economy + year fixed effects included in every model. Wild-cluster bootstrap p-values in parentheses (9,999 replications, Rademacher weights, clustered by economy).

4.2 A suggestive conditional convexity (Proposition 2)

Once technological capability and digital adoption are held constant (M3), the curvature turns more strongly positive and **marginally significant** ($\beta_2 = +1.051$, wild-bootstrap $p = .056$), with a negative linear term ($\beta_1 = -0.852$, $p = .024$). The implied shape is, tentatively, **convex (U-shaped)**: productivity declines with export intensity across the low-to-moderate range and would turn up only beyond an interior *minimum* in the high-FSTS region that few island firms reach. We read this as *suggestive* of a mirror image of the mainland inverted-U rather than a confirmed reversal, given the marginal significance and the small number of clusters.

Figure 1. The inverted-U dissolves at the periphery: mainland (n) vs Pacific SIDS (curvature not significant)



Substantively, if the convex pattern holds, the dominant mass of SIDS firms—almost all at low-to-moderate FSTS—would sit on the **descending limb**, where additional export intensity is associated with lower productivity: the empirical signature consistent with the FIP mechanism.

4.3 Capability raises the level but does not restore the curve (Proposition 3)

Technological capability carries a positive, robust level effect (TCI: $\beta = +0.064$, wild-bootstrap $p = .036$); digital adoption is positive but weaker and non-significant (DAI: $\beta = +0.157$, $p = .102$). Neither restores a concave inverted-U—indeed, conditioning on capability sharpens the *convex* pattern. Capability thus operates as a productivity level-shifter that cannot substitute for the absent structural prerequisites, consistent with Proposition 3.

4.4 Transparency: sensitivity of any strong negative estimate

We report openly that the *magnitude* of any strong negative I–P estimate in this setting is sensitive to which island economies enter the sample. A more restrictive build limited to the few economies with complete pre-2018 control data yields a large, significant negative slope, whereas the full seven-economy sample yields the null-to-convex pattern reported above; with only three clusters, cluster inference is unreliable in the restrictive build. We therefore anchor

our claims on the full sample with wild-bootstrap inference and present FIP as a theoretical limiting case rather than a precisely estimated effect. This transparency is itself a methodological contribution to firm-level research on data-scarce micro-economies.

5. Discussion

5.1 Bounding the inverted-U paradigm

Our central result is that the inverted-U I–P relationship does not extend to the Pacific periphery. The shape that organizes the I–P literature dissolves, with suggestive evidence that—conditional on capability—the relationship leans convex rather than inverting cleanly. This is not a failure of measurement but a **boundary condition**: the inverted-U presupposes a viable domestic market, manageable trade costs, and institutional support, and where these are jointly absent the relationship the paradigm predicts simply does not obtain. The finding reframes a long line of work (Hitt et al., 1997; Lu & Beamish, 2004) as *conditionally* rather than universally applicable, and it offers a structural explanation for the weak and unstable estimates that large-sample replications have reported (Berry & Kaul, 2016; Pisani et al., 2020): pooling prerequisite-rich and prerequisite-poor contexts averages incompatible functional forms.

5.2 The contribution of FIP as a delimited concept

By naming and delimiting the Forced Internationalization Penalty, we give the field a label for the limiting case at the extreme periphery and a set of distinguishing criteria separating it from phase-one S-curve costs, the liability of foreignness, and necessity entrepreneurship. The construct’s value lies in marking where universal internationalization theories stop, independent of the precise coefficient one estimates—an instance of theory development through the specification of scope conditions.

5.3 Policy implications

For development policy, the result cautions against export-intensification as a stand-alone growth lever in microstates. If the binding constraints are domestic-market viability, trade costs, and institutions, then promoting export intensity without addressing those prerequisites pushes firms further onto the descending limb. Complementary structural interventions—regional market integration to relax the domestic-scale constraint, trade-cost reduction, and targeted institutional development—are prior conditions for export-led growth to deliver productivity gains. This aligns with evidence that development progress in SIDS has been the slowest on record (Mahler, Serajuddin, Wadhwa, & Yonzan, 2026).

5.4 Limitations and future research

Three limitations bound the findings. First, the design is cross-sectional repeated cross-sections with few clusters; we mitigate few-cluster over-rejection with wild bootstrap, but causal identification awaits panel data across more island economies. Second, the digital-adoption measure is Tier-1 (website) only, the comparably available indicator across SIDS waves; richer digital capability cannot be assessed. Third, generalization beyond the Pacific is untested—Caribbean and Indian-Ocean SIDS are natural extension contexts that would also raise the cluster count needed for sharper inference. A definitive test of FIP as an estimated effect, as opposed to a bounding concept, requires more complete firm-level data across a larger set of microstates.

6. Conclusion

The inverted-U is the workhorse functional form of the internationalization–performance literature, but its generating logic presupposes structural prerequisites that the world’s most peripheral economies lack. Using firm-level data from seven Pacific island states, we show that the

inverted-U dissolves at this periphery, with suggestive evidence that, conditional on capability, the relationship leans convex—a penalty across the export-intensity range where island firms actually operate. We name the limiting case the Forced Internationalization Penalty and position it as a scope condition of universal internationalization theory. The paradigm is conditional, not universal: where the domestic market is too small, trade costs too high, and institutions too thin, intensifying internationalization is not a path to performance but a structural burden.

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